

The model examined the relationship between fiscal policy (tax revenue and government consumption) and GDP. Preliminary testing that includes the three independent variables such as tax revenue, government consumption, and foreign direct investments showed the presence of multicollinearity as indicated by an inflated value of the variance inflation factor for tax revenue and government consumption. The results show a strong linear relationship between the two independent variables. To address this issue, the two variables were combined into a single composite variable. This resolves the multicollinearity issue while preserving the data and theoretical integrity of the model.

After resolving the issue of multicollinearity the preliminary model also showed issues with autocorrelation using the Breusch-Godfrey Serial Correlation LM Test which shows results that confirms first-order correlation. To correct this a dynamic regression model was used. The autoregressive distributed Lag model was applied to capture economic effects that often unfold over time. This adjustment corrected the problem with autocorrelation and it also enhanced the model's reliability and interpretative accuracy by integrating the concept of "carry-over effect" of GDP.